

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7 HT	(a)	(i)		Indicative content Early on He and H₂ escaped from atmosphere. Water vapour present in early atmosphere (from volcanoes) condensed to form oceans. (As the earth cooled) the % of carbon dioxide has decreased to a fraction of 1%. This is due to: photosynthesis in green plants, using up carbon dioxide and releasing oxygen; More CO₂ locked up by animals evolving shells; subsequent formation of limestone; Carbon lock up in fossil fuels from the remains of simple marine organisms. Ammonia decomposed by a reaction with oxygen forming nitrogen the most abundant gas in the atmosphere. Reference to time 4.5-4.0 billion years ago and present day. 5– 6 marks Candidates can give a comprehensive description of the changing composition of the atmosphere. They recognise the major changes in all gases. They give a comprehensive explanation of why these changes have occurred from early on to present day. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3 – 4 marks Candidates can quantitatively (time or %) describe some changes in the atmosphere. OR Candidates can give some explanation or reasons for the changes in water vapour, carbon dioxide or oxygen. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology</i>		2	4	6		